Auss Abbood

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EDUCATION

University of Cambridge, Cambridge, UK — Oct. 2024 – present
PhD in Computation, Cognition and Language

Osnabrück University, Osnabrück, Germany — Oct. 2016 – May 2019
MSc in Cognitive Science — Grade: 1.1, on a scale from 1 (excellent) to 5 (bad)
Specialised in artificial intelligence, neuroinformatics, and neuroscience. — With distinction

Goethe University Frankfurt, Frankfurt am Main, Germany — Oct. 2013 – Sep. 2016
BSc in Biological Sciences — Grade: 1.6, on a scale from 1 (excellent) to 5 (bad)
Specialised in neuroscience.

PUBLICATIONS

- Hicketier, A., Bach, M., Oedi, P., Ullrich, A., **Abbood, A.** (2026). Ensemble-labeling of infectious disease time series to evaluate early warning systems. *Infectious Disease Modelling, 11(3)*, 823–839.
- **Abbood, A.**, Meng, Z., Collier, N. (2025). *Time* to Revisit Exact Match. *Findings of the Association for Computational Linguistics: EMNLP 2025*, 11903–11926.
- Zhang, M., **Abbood, A.**, Meng, Z., Collier, N. (2025). The Power of Simplicity in LLM-Based Event Forecasting. *Proceedings of the 1st Workshop for Research on Agent Language Models (REALM 2025)*, 454–470.
- Bhatia, S., . . . , **Abbood, A.**, . . . , Cori, A. (2023). Lessons from COVID-19 for rescalable data collection. *The Lancet Infectious Diseases, 23(9)*, e383–e388.
- **Abbood, A.**, Ullrich, A., Denkel, L. A. (2023). Understanding COVID-19 reporting behaviour to support political decision-making: a retrospective cross-sectional study of COVID-19 data reported to WHO. *BMJ Open, 13(1)*, e061717.
- **Abbood, A.**, Ghozzi, S. (2022). General Framework for Evaluating Outbreak Prediction, Detection, and Annotation Algorithms. *medRxiv* [Preprint].
- Oala, L., . . . , **Abbood, A.**, . . . , Wiegand, T. (2021). Machine Learning for Health: Algorithm Auditing & Quality Control. *Journal of Medical Systems, 45*, article 105.
- **Abbood, A.**, Ullrich, A., Busche, R., Ghozzi, S. (2020). EventEpi—A natural language processing framework for event-based surveillance. *PLOS Computational Biology, 16(11)*, e1008277.

EXPERIENCE

UK AI Security Institute — London, UK
Research Scientist (CAM-DTP Innovation Fellowship) — Jul. 2026 – Oct. 2026

- Evaluated AI agents' research capabilities on long-horizon AI R&D tasks using Transect, AISI's open-source transcript analysis toolkit.
- Identified measurable indicators of research skill and research taste through a literature review of human research practices and analysis of agents' behaviour.

Robert Koch Institute — Berlin, Germany
Data Scientist and Project Lead — Oct. 2018 – Sep. 2024

- Built a risk assessment pipeline for COVID-19 by combining tabular and text data, applying self-developed risk metrics, and predicting short-term risk, saving 1 h of manual work daily.
- Co-developed intensivregister.de (ICU surveillance) working on data engineering, visualisation, and on-demand data analyses to support ICU management in Germany, serving up to several million visits per day.
- Automated key information extraction and relevance scoring for public health surveillance to potentially save epidemiologists 30 minutes of daily work.
- Built collaborations with partners from the public and private sector from more than a dozen countries and with organisations such as the European Centre for Disease Prevention and Control, Africa Centres for Disease Control and Prevention, and the World Health Organization Hub.
- Led and supervised applied data-science work, including supervision of student assistants and data scientists.

Communintelligence — Münster, Germany
Co-founder and Data Scientist — Oct. 2017 – Jan. 2019

- Provided data science consultancy for a public utility company, predicting bus delays with a two-minute mean error using random forests.

PROJECTS

Public Health Intelligence Capacity Building and Innovation Berlin, Germany
Project Link: PHI-CBI Jan. 2023 – Oct. 2024

- Led technical work on Epidemic Intelligence from Open Sources (EIOS), identifying surveillance workflow bottlenecks through user interviews and data analyses.
- Evaluated recommender systems, anomaly detection, and text classification to improve article relevance.
- Co-developed a pipeline to improve the signal-to-noise ratio of the EIOS data stream.

Data and AI-supported early warning system to stabilise the German economy Berlin, Germany
Project Link: DAKI-FWS Dec. 2021 – Oct. 2024

- Co-led early-warning research using data from Germany's notifiable disease surveillance system, wastewatermonitoring, and self-reported symptom data.
- Built a forecast hub with specialised evaluation metrics and developed a labelling algorithm for COVID-19 time series to evaluate AI-based early-warning models.

Natural Language Processing for Event-based Surveillance with Africa CDC Berlin, Germany
Project Link: NaLaA May 2021 – Jan. 2024

- Led NLP-based Twitter surveillance with Africa CDC, developing a Sesotho geotagging model and integrating surveillance into its event management system.
- Co-organised a workshop on event-based surveillance methods for the Southern Africa RCC.

The Focus Group on Artificial Intelligence for Health by ITU and WHO
Project Link: FG-AI4H Nov. 2019 – Aug. 2023

- Co-led the Topic Group on Outbreaks and AI test specifications, defining best practices for developing and benchmarking outbreak-detection algorithms.
- Developed a benchmarking setup combining qualitative and quantitative evaluation, and contributed guidance on safe AI use.

CONFERENCE PRESENTATIONS

- Zhang, M., **Abbood, A.**, Meng, Z., Collier, N. (2025). The Power of Simplicity in LLM-Based Event Forecasting. REALM – Workshop at ACL.
- Presented talks and posters at AMLD, ESCAIDE, DGEpi, EPIDEMICS 8, and a NeurIPS workshop (2019–2021).

GRANTS AND FUNDED PROJECTS

- PHI-CBI: Contributed to successful acquisition of €1.6M for public health intelligence capacity building and innovation, 2023–2024.
- DAKI-FWS: Contributed to successful acquisition of €860K for data- and AI-supported early-warning systems, 2021–2024.
- NaLaA: Led acquisition of €590K for NLP-based event-based surveillance with Africa CDC, 2021–2024.

TEACHING AND SUPERVISION

- Teaching assistant, Li18: Computational Linguistics, University of Cambridge.
- Lecture on outbreak detection for public health surveillance at the Robert Koch Institute.
- Training of trainers on data science in epidemiology for epidemiologists in Uzbekistan.
- Supervised five data scientists and one student assistant; co-supervised one master's thesis.

AWARDS AND SCHOLARSHIPS

- CAM-DTP PhD studentship funded by the ESRC.
- Winner, MOOD EU Horizon 2020 summer school hackathon, for an ensemble ML model predicting tick prevalence from spatio-temporal earth observation, vegetation, and weather data, 2023.
- Rosa-Luxemburg-Foundation Scholarship, 2018.
- Second place, hack4health, for a Google Trends-based early-warning system and chatbot for syndromic surveillance, 2018.
- Winner, MÜNSTERHACK, for AR-based public transport information and ML-based bus-delay prediction, 2017.
- Winner, Hasso-Plattner-Institute Hackathon, for a vaccine-consultation chatbot, 2017.

TECHNICAL SKILLS

- **Programming languages:** Python, R, Java, SQL, Bash
- **Tools and libraries:** Altair, BeautifulSoup, gensim, Flair, Git, Luigi, Matplotlib, NLTK, NumPy, pandas, Plotly, PyTorch, SciPy, Selenium, scikit-learn, spaCy, statsmodels, TensorFlow
- **Languages:** German (fluent), English (fluent), Arabic (good)